

## Education

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|------------------------------------------------|---------------------------------------|-----------|------|
| <b>BTech</b> - Robotics and Autonomous Systems | Plaksha University, Mohali            | 8.98 / 10 | 2027 |
| <b>12th</b> - CBSE                             | Global Indian Intl. School, Abu Dhabi | 95%       | 2023 |
| <b>10th</b> - CBSE                             | Global Indian Intl. School, Abu Dhabi | 92.4%     | 2021 |

## Experience

**Research Intern, UR Rao Satellite Centre, Indian Space Research Organisation (ISRO)** Jun 2026 - Jul 2026

- Built a **robot and simulator agnostic training framework** to design hierarchical locomotion controllers for sim-to-sim/real learning.
- Designed **hierarchical RL controllers** combining a learned high-level foot placement policy with optimisation-based **MPC and QP** for joint control.
- Benchmarked learnt latent **reduced-order models** against first-principles models (LIP, ALIP, SRB) through max speed and disturbance metrics.
- Trained foot-placement policies tracked via a task-space QP, achieving **<0.05 m/s velocity tracking error** and surviving **1.3m/s lateral impulses**.

**Student Researcher, Dixon IoT Lab, Plaksha University** Aug 2025 - Jun 2026

- Air Quality Analysis of North India:**
  - Scraped over **30 million datapoints of live air quality data** from across north India, at a frequency of 10 minutes for over 4 months.
  - Developed a novel encoder-decoder model to **analyze spatio-temporal air quality data** and identify regional pollution trends over time.
  - Conducted a **divergence analysis** of pollution motion across North India by integrating wind data to find pollution sources and sinks.
- Air Quality Mapping of Chandigarh Tricity Area:**
  - Designed and deployed low cost **IoT Air Quality Monitors** to create a dense Air Quality Map of the Chandigarh Tricity area.
  - Developed a custom **C++ sensor library** supporting 12 sensors, and 3 communication protocols with dynamic runtime sensor initialization.
  - Built an interactive web dashboard to visualize local air quality analytics for deployment on the **Chandigarh Citizens Foundation** website

**Summer Reserach Intern, Prof. Sunita Chauhan, Plaksha University** Jun 2025 - Aug 2025

- Developed the **UGV subsystem** of a collaborative UAV-UGV precision agriculture platform for autonomous crop monitoring.
- Built a **ROS 2/Gazebo digital twin** of a UGV and 4-DoF manipulator payload and tuned appropriate ROS Controllers.
- Deployed a **full navigation stack** - SLAM, localization, path planning - with **LiDAR and UAV imagery maps** on a testbed UGV.

## Publications

Kapoor A, **Dayma S**, Narula M, Srinivasan S, *A Framework for High-Resolution Air Quality Monitoring for the Chandigarh Tricity Region*, 15th International Conference on Air Quality - Science and Application.

## Projects

**Inverse PINN Modelling of Flow in a Compliant Tube** | Prof. Shashikant Pawar Aug 2026 - Present

- Analyzing flow–structure interactions in collapsible tubes to study pressure and velocity fields in **venous blood flow**.
- Developing fluid–structure interaction (FSI) models and experimental setup for tube deformation and flow measurements.
- Developing a physics-informed inverse model using PINNs and CFD pipelines for validating experimental measurements.

**FOC Motor Control** | Prof. Srikanth Srinivasan Apr 2026 - May 2026

- Implemented **field oriented control** for a gimbal BLDC motor on **STM32** enabling **torque control** mode.
- Generated center aligned **sinusoidal PWM** via **Clarke and Park transforms**, with quadrature encoder feedback and startup rotor alignment.
- Designed **inline current sensing** and timer-triggered ADC sampling, enabling 90%+ efficiency and near 0 torque ripple.

**Comparison of Neural Network and RL based Controls** | Prof. Sandeep Manjanna Feb 2026 - May 2026

- Comparison of **physics informed neural network (PINN)** based HJB-PDE solvers and RL algorithms for the inverted pendulum control problem.
- Constructed inverted pendulum system for hardware testing of policies, achieving **<10 degrees of tracking error** with naive PD position control.
- Determined PINN based controllers yield more optimal solutions with **<0.002 mean reward deviation from traj opt.** benchmark.

**Digital Guitar Pedal** | Prof. Srikanth Srinivasan Feb 2026 - May 2026

- Engineered a custom **16-bit, 44.1kHz digital effects pedal** for the guitar from scratch, alongside an enclosure and full UI Interface.
- Designed **single supply biasing op-amp circuits** for pickup signal level and impedance matching with downstream ADC.
- Implemented **real-time, low-latency audio processing** on the Teensy 4.1, including a 10-stage tone stack, distortion, delay, reverb, etc.

**Optimal Control using PINNs** | Suraj Kumar, URSC, ISRO Sep 2025 - Dec 2025

- Developed physics informed neural networks to solve infinite horizon **optimal control problems** such as **inverted pendulum** and **cart pole**.
- Leveraged theory of functional connections and extreme learning machine to **solve Hamilton-Jacobi-Bellman PDE**.
- Proposed XTFC-Unfreeze** model with **LQR initialization** showcasing improved non-linear control and stability for standard systems.

**Lighter Than Air Systems** | Prof. Rajkumar Pant Jan 2025 - May 2025

- Designed and fabricated a number of lighter than air systems such as **aerostat, airship envelopes, recovery devices, and launchers**.
- Engineered and deployed a **glider launch mechanism on an aerostat**, achieving a **30m+ glide from a 10m altitude**.
- Designed and validated an **emergency rapid deflation system** for airships, enabling both payload and envelope material recovery.

## Skills

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**Programming Languages:** Python, C, C++, SQL, Bash, Assembly (ARM Cortex-M, MIPS)  
**Environments:** ROS2, Gazebo Simulator, MuJoCo, MJX  
**Math & Optimisation:** Linear Algebra, Dynamics, Trajectory Optimisation, Model Predictive Control  
**Other Skills:** Git,  $\LaTeX$ , CAD (Fusion 360, Onshape), CFD (COMSOL, SimScale, OpenFOAM)

## Relevant Coursework

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|----------------------------------------------------------------------------|-----------|
| RO2001: System Dynamics and Controls   Prof. Sunita Chauhan                | Grade: A+ |
| RO3003: Control Autonomy Planning and Navigation   Prof. Shashank Tamaskar | Grade: A+ |
| RO3101: Robot Motion, Simulation, and Optimisation   Prof. Andy Ruina      | Grade: A  |
| CP3001: Embedded Systems   Prof. Srikanth Srinivasan                       | Grade: A  |
| CP2002: Signals and Systems   Prof. Vivek Deulkar                          | Grade: A  |
| ME3001: Engineering Mechanics   Prof. Shashikant Pawar                     | Grade: A  |
| ME3002: Fluid Mechanics   Prof. Shashikant Pawar                           | Grade: A  |

## Teaching

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| <b>Student Tutor, FM214: Intelligent Machines   Plaksha University</b>                                                                                                                                                                                                                                                                                                                                                                                                                   | Aug 2026 - Present  |
| <ul style="list-style-type: none"><li>Supported instruction for sophomore robotics course with <b>230+ students</b>, under Prof. Andy Ruina and Prof. Sandeep Manjanna.</li><li>Assisted in conducting tutorials and laboratory sessions, related to mobile robots, estimation, planning, and control.</li></ul>                                                                                                                                                                         |                     |
| <b>Student Tutor, FM206: Mechatronics   Plaksha University</b>                                                                                                                                                                                                                                                                                                                                                                                                                           | Aug 2025 - Dec 2025 |
| <ul style="list-style-type: none"><li>Supported instruction for a Semester 3 introductory robotics course with <b>180+ students</b>, under Prof. Andy Ruina.</li><li><b>Designed demonstrations</b> to teach concepts like ADCs, circuit analysis, DC motors, servo motors, PWM, odometry and, forward kinematics.</li><li>Assisted in <b>conducting tutorials and laboratory sessions</b>, ensuring safe practices, debugging hardware, and helping students build intuition.</li></ul> |                     |

## Positions of Responsibility

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| <b>Cohort Representative, Robotics and Autonomous Systems   Plaksha University</b>                                                                                                                                                                                                                                    | Jul 2026 - Present  |
| <ul style="list-style-type: none"><li>Represented RAS cohort, facilitating communication between 20+ faculty members and the student body.</li><li>Bridging student-faculty communication on course policies resolving student academic challenges through continuous feedback.</li></ul>                             |                     |
| <b>Co-organiser, TEDxPlakshaUniversity</b>                                                                                                                                                                                                                                                                            | Jan 2024 - Apr 2024 |
| <ul style="list-style-type: none"><li>Launched the <b>first TEDx event</b> at Plaksha University, directing a <b>10+ member team</b> across speaker curation, production and outreach.</li><li>Curated <b>5 speakers</b> across disciplines; recorded talks have since drawn <b>100k+ views</b> on YouTube.</li></ul> |                     |

## Achievements

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- Abu Dhabi Student Golden Visa:** Long-term UAE residence visa awarded for academic excellence.
- Merit Scholarship, Plaksha University:** Recognized for academic excellence and extracurricular contributions.
- Academic Excellence Award** from Rochester Institute of Technology, Dubai.